

FINAL_RESEARCH_REPORT.md

DSN Bootcamp Qualification Hackathon 2026 ML Track

Final Research Report

Report Date: 2026-09-22
Project Root: C:\Users\user\Downloads\DSN_HACKATHON_2026

1. Competition Objective

Predict `total_sales` for products across stores using historical sales data. The competition uses RMSE as the evaluation metric.

2. Dataset Summary

Split	Rows	Features	Target
Train	6,818	19 (after preprocessing)	<code>total_sales</code>
Test	1,705	19 (same schema)	—

Categorical features (6): `product_code`, `store_code`, `product_category_clean`, `store_format`, `store_location_tier`, `fat_content_clean`

Numeric features (13): `product_weight_kg_imputed`, `shelf_visibility`, `product_price`, `store_age_years`, `missing_weight`, `missing_size`, `visibility_zero`, `log_price`, `log_weight`, `log_visibility`, `store_size_encoded`, `tier_encoded`, `fat_encoded`

3. Canonical Baseline

- **Model:** Optimized blend of 5 models with frozen weights
- **Public Score:** 1073.18868
- **SHA256:** 55f346a2d7a94e10f83cf5e393013a113d74210ab128ce95dd89501277c354e4
- **Blend weights:**
 - A_direct: 0.3150392234808109
 - B_store_norm: 0.5635523048051151
 - C_two_stage: 0.08326192124539074
 - D_ridge: 0.038146550468683295
 - E_catboost_ridge: 0.0

4. Research Chronology

Phase	Topic	Result
PHASE4	Data Forensics	Validated
PHASE5	Baseline Establishment	Validated
PHASE6	Tail Analysis	Rejected
PHASE7	Store Representation	Inconclusive
PHASE8	Product Representation	Inconclusive
PHASE9	Feature Engineering	Inconclusive
PHASE10	Cross-Phase Synthesis	Validated
PHASE11	E3 CTR Complexity	Rejected (public)
PHASE11C	Production Config	Validated
PHASE12	E3 Generalization	Invalidated
PHASE13	Validation Comparison	Validated
PHASE14	Product-Store Structure	Closed
PHASE15	Pair Holdout	Invalidated (bug)
PHASE16	Validation Ambiguity	Validated
PHASE17	Data Forensics	Invalidated
PHASE17A	Phase 17 Audit	Invalidated claims
PHASE17B	Research Reopening	Outcome C
PHASE18	Ensemble Audit	Outcome C
PHASE19	Canonical Verification	Validated
PHASE23	Consensus Robustness	Validated

5. Major Rejected Branches

- **E3 (max_ctr_complexity=3):** Public RMSE 1084.00044 — rejected by public leaderboard.
- **VIS_RATIO:** Public RMSE 1087.75078 — rejected.
- **E+G:** Public RMSE 1094.67417 — rejected.
- **Phase 15 Product-Store Holdout 2570 RMSE:** Invalidated due to RMSE calculation bug.
- **Phase 17 model identity error:** P17-S2-02/03 were CatBoost predictions mislabeled as LGBM/XGBoost.
- **Phase 18 ensemble blends:** All 48 blends worse than CatBoost; residual correlations 0.95–0.98.

6. Phase 23 Findings

Phase 23 conducted a consensus robustness search across multiple validation protocols (SKF, PMH, TSM) and seeds. Key findings:

- Canonical control values reproduced exactly for seeds 42, 123, 999.
- No globally dominant feature combination found across all 9 validation cells.
- E+G is NOT globally dominant (1/9 cells worsened).
- Simplest evidence-supported candidate: E_alone (1 feature, 7/9 improved).
- All Stage 3 deployment safety checks passed (18/18).

7. Public Leaderboard Validation

Candidate	Public Score	Status
OPTION_1_CANONICAL	1073.18868	FROZEN
OPTION_2_E3	1084.00044	RETAINED ALTERNATE
VIS_RATIO	1087.75078	REJECTED
E	1090.57216	REJECTED
E+G	1094.67417	REJECTED

Private leaderboard performance: UNKNOWN

8. Final Submission Options

OPTION_1_CANONICAL is the primary final submission.

OPTION_2_E3_HEDGE is retained as the only alternate final-submission option.

No other candidates are final submission options.

9. Reproducibility Methodology

- Canonical: Verified byte-identical via `CODE/final/run_notebook_repro.py`
- E3: Verified byte-identical across 2 clean processes with CatBoost 1.2.10
- All hashes computed with SHA256
- All schemas validated programmatically

10. Known Limitations

1. E3 requires CatBoost 1.2.10 exactly for byte-identical reproduction
2. E3 script must run from ARCHIVE/REPORTS/ for relative paths
3. Private leaderboard performance is unknown for both options
4. No new experiments or submissions are authorized by this package

11. Exact Hashes

File	SHA256
OPTION_1_CANONICAL/FINAL_SUBMISSION.csv	55f346a2d7a94e10f83cf5e393013a113d74210ab128ce95dd89501277c354e4
OPTION_2_E3/PHASE11C_SHADOW_SUBMISSION_E3.csv	8805c96e3411acc154429e6b8376446e8bfdf75ba453e1c05ed534373f4f962

12. Final Frozen State

- Canonical submission: FROZEN, verified unchanged
 - E3 submission: RETAINED, verified unchanged
 - No new submissions authorized
 - Research closed
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Report generated as part of FINAL COMPETITION RELEASE / REPRODUCIBILITY WAR ROOM.